MA6094 Capstone Brief — Harbourline Community Hub

Oct 10, 2026 · @Spurthi

At a glance

Your team designs, builds and defends an AI-driven project management system for one deliberately incomplete project: the Harbourline Community Hub Refurbishment. The system must integrate planning, risk and reporting agent modules, write into a real PM tool behind a human checkpoint, and show evidence that it works.

Item	Detail
Weight	40% of the course grade (team and individual). The separate 5% individual reflection is written about this project.
ILOs assessed	2, 3, 4, 5, 6, 7, 8
Team size	5–6 students
Launch	Week 9
Build sprint	Week 12, in class
Submission (the only checkpoint)	Group A by 23:59 Wed 11 Nov 2026 · Group B by 23:59 Thu 12 Nov 2026
Final presentation	Week 13, Fri 13 Nov 2026: live demo and panel Q&A
You hand in	Design document, working system, evaluation report, governance log, presentation, peer evaluation
Your sponsor	Daniel Koh, Director, Estate Development, Harbourline Town Council, played by the instructor. Send questions at any time before you submit; answers are posted to every team.

The brief leaves real gaps on purpose. A strong system surfaces them and asks; a weak one invents answers and builds on them. How your system handles what the brief does not say is assessed throughout.

The project brief

The brief below is written as the sponsor handed it over: short, confident and incomplete. Harbourline Town Council and every organisation in it are fictional. Treat the text as your only source of truth until the sponsor answers your questions.

From: Daniel Koh, Director, Estate Development, Harbourline Town Council **Subject:** Harbourline Community Hub Refurbishment — project brief

Background. Harbourline Community Hub is a four-storey building completed in 1994, about 9,800 m² gross floor area. It houses a sports hall, multi-purpose rooms, a tuition centre and a small food court, all on leases or licences. Usage has fallen, the M&E plant is at end of life, and residents have asked for childcare and senior services closer to home.

Objectives. Refurbish the Hub into an integrated community facility that serves residents of all ages, improves accessibility and energy performance, and becomes the estate's flagship shared space.

Scope. The refurbished Hub will include:

- an upgraded multi-purpose sports hall
- a new early childhood centre for about 100 children, run by our selected operator, Little Harbour Learning
- an active ageing centre, operator to be appointed
- a community kitchen and food court refresh
- a rooftop community garden
- full barrier-free access, new lifts and replacement of M&E systems

Budget and time. Approved budget is S$18.5 million. Sixty per cent is funded by a central grant released in tranches against milestones; grant conditions will follow. Works are to complete within 22 months, with the Hub fully open by the fourth quarter of 2028.

Delivery. We intend a single design-and-build contract. The Hub must remain open to residents throughout the works. Residents will be consulted on the design. Little Harbour Learning wishes to begin enrolment in month 18.

Reporting. The Estate Development Committee meets weekly and expects a short status update. The grant body requires a monthly progress report. Our current tracking is in spreadsheets.

Success. A fully operational Hub, delivered on budget, with high resident satisfaction.

Read it twice before you build anything. Several statements cannot all be true at once, and several decisions a planner needs are missing entirely. List them, and send the ones only the sponsor can answer as early as you can.

What your system must do

Your system needs six modules plus a governance layer around all of them. Each module draws on a block of the course, so nothing here is new; what is new is making them work as one system. The minimum column is required to pass that criterion; the stretch column is where higher bands are earned.

Module	Minimum	Stretch	Course weeks	ILO
1. Planning	Scope statement with Questions for the Sponsor; WBS to Level 3 with dictionary; milestone plan with dependencies, in relative time until the sponsor gives dates	Every WBS element traceable to a scope line; anything not traceable is flagged, not kept	2–3	2
2. Tool integration	WBS and milestones written into Jira (or Asana / MS Planner) behind a human confirmation; system reads the board back	Stable external keys from WBS codes, so a second run updates rather than duplicates	4–5	6
3. Memory and retrieval	Answers questions over the data pack and cites the document and line it used	Says NOT FOUND when the pack is silent; carries decisions forward between sessions	9–10	3
4. Risk and what-if	Scored risk register with owners and triggers, each risk linked to the work it threatens; one what-if with schedule and cost impact and stated assumptions	Three what-ifs compared; risks re-scored automatically when the plan changes	11	3, 7

Module	Minimum	Stretch	Course weeks	ILO
5. Reporting with a critic	Weekly committee update and monthly grant report from the same facts; a critic agent checks both before a human sees them, with a stop condition	Critic findings logged; you can show a report the critic stopped	6–7	6, 8
6. Change handling	Responds to the Week 12 change event: detects it, shows what in the plan, register and reports is now stale, drafts actions for a human	Stale outputs refresh without being re-run by hand	12	2, 7
Governance layer	Named human checkpoints; audit log of every agent write; agents never delete or close work items	A bias and fairness check on the resident-consultation summary	10	5

Two integration rules apply to the whole system. At least one hand-off between modules must run without a person copying output across. And every place a person must approve something has to be designed in, not discovered.

Constraints, tools and guardrails

You may build with code or without it; the grade rewards the system's reasoning and control, not the amount of code. Every required feature must run on free tiers. A paid feature may appear in your demo only if the same requirement is also met on a free route.

Use for	Permitted	Notes
Agents and prompts	ChatGPT, Claude, Gemini, Google AI Studio	Keep Role and Context fixed per agent, as in Weeks 2–3
No-code chaining	Google Opal, n8n, Zapier	Opal cannot loop or branch: say how you work around that

Use for	Permitted	Notes
Code orchestration	Python with the OpenAI Agents SDK, CrewAI or LangChain	Optional. Expected for the stretch column of Module 6
PM tool	Jira Cloud Free (recommended), Asana, MS Planner	Your team's own site, never a work instance
Data	The provided data pack only	No employer or client material in any prompt

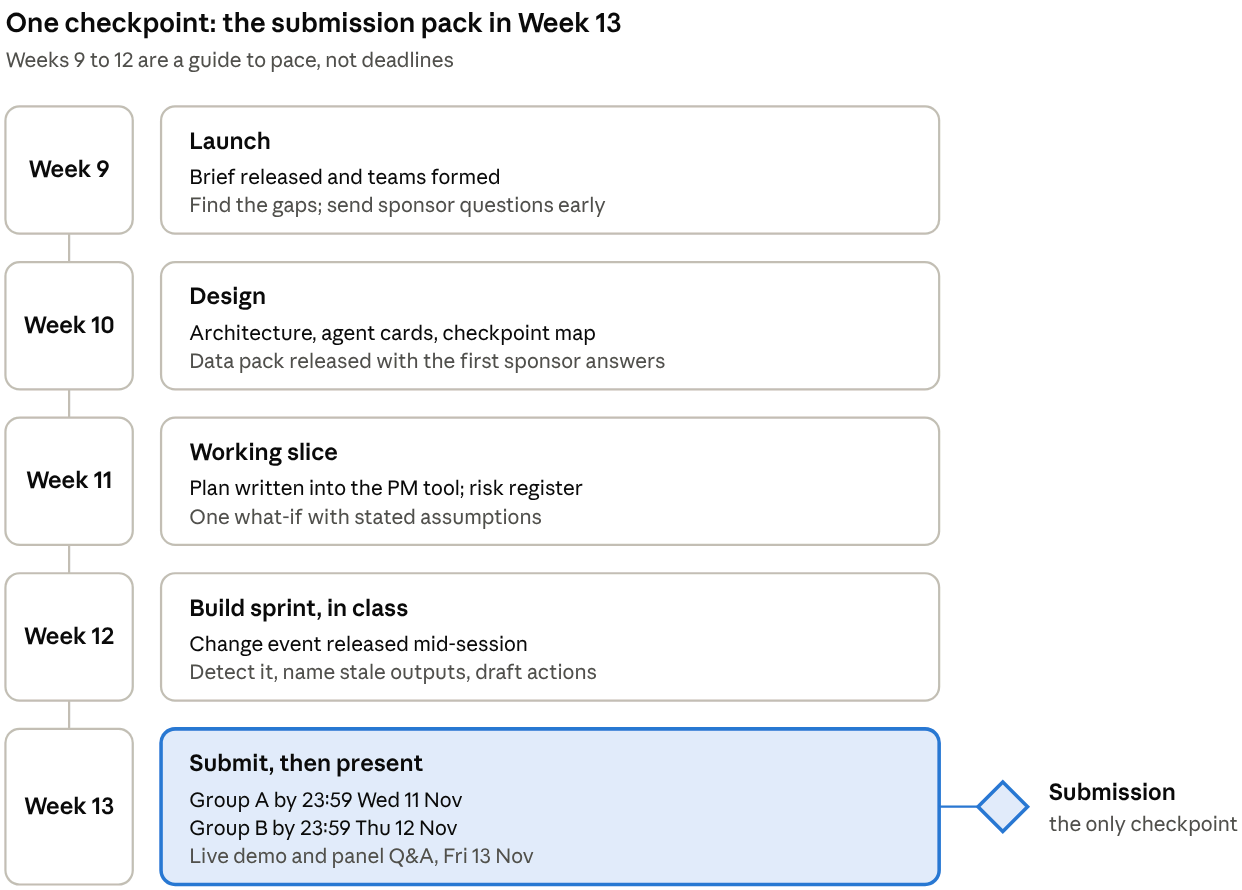
Do not build on OpenAI Agent Builder, which its vendor has announced will shut down on 30 November 2026, or on ChatGPT agent mode, which has been withdrawn. Check any tool you depend on again in Week 12; vendor change is itself a risk you should log.

Guardrails that apply to every team:

- One sandbox project in your own PM tool site; label every agent-created item `ai-generated`.
- Agents may create, update and comment. Moving work to Done, closing or deleting is a human act.
- API tokens are personal. They never appear in shared documents, slides, screenshots or recorded prompts.
- Start read-only; add write access only after the read path works and is logged.

Milestones

There is one checkpoint: the submission pack, due by 23:59 on Wed 11 Nov 2026 for Group A and Thu 12 Nov 2026 for Group B. The weeks before it are yours to plan; the guide below shows where a team on track should be each week.



capstone roadmap · Weeks 9–13, four checkpoints

Sponsor channels. Send questions to the sponsor at any time before you submit, through NTULearn. There is no question deadline, but early questions get answers in time to use them. Answers are posted to every team, and anything the sponsor declines to answer stays an open assumption in your register.

Deliverables

The submission pack is the only checkpoint. Group A submits by 23:59 on Wed 11 Nov 2026 and Group B by 23:59 on Thu 12 Nov 2026. Everything is submitted as files or links; nothing is graded from memory of the demo alone.

#	Deliverable	What it must contain	Limit
1	Design document	Architecture diagram; one agent card per agent (role, inputs, outputs, tools, stop condition); human checkpoint map; data contract with your PM tool; assumption register with the sponsor's answers	8 pages plus appendices
2	Working system	A link or repository the panel can open, plus a recorded end-to-end run used if the live demo fails	Recording 10 min
3	Evaluation report	Test set of at least 20 cases, including the brief's gaps and the Week 12 change; results on the five measures below	4 pages
4	Governance log	Exported audit trail of agent writes; every checkpoint with who approved; bias and fairness check; incidents and what you changed	Export plus 2 pages
5	Presentation	Slides for the Week 13 panel	12 min plus Q&A
6	Peer evaluation	Each member rates each teammate's contribution, submitted individually and confidentially	One form each

The five evaluation measures every team reports, so results can be compared across teams:

1. **Gap recall** — how many of the brief's open questions your system surfaced unprompted.
2. **Invention rate** — share of sampled claims in outputs that cannot be traced to the brief, the data pack or a sponsor answer.
3. **Re-run stability** — what changes, and what duplicates, when the same input runs twice.
4. **Critic catch rate** — faults you planted in a draft report that the critic found.
5. **Cost and time per full run** — tokens or credits and minutes, as in Week 7.

The individual reflection (5%, a separate component) is written about this project and due by 23:59 on Fri 20 Nov 2026.

Assessment

The capstone is worth 40% and has three components, as filed in the course outline: system build, live presentation and demonstration with panel Q&A, and peer evaluation. The 40% is divided as below.

Component	Weight	Graded on	Team or individual
System build	20%	Deliverables 1–4	Team
Live presentation, demo and panel Q&A	15%	Deliverable 5 and the live session	Team, with individual answers
Peer evaluation	5%	Deliverable 6	Individual

What the panel looks for, in the five areas the course outline names:

Area	A strong team shows	ILO
System integration	Modules hand off to each other; the PM tool reflects the plan; stale outputs are caught after a change	2, 6
Technical implementation	The system runs end to end on free tiers; re-runs do not duplicate; failures are visible, not silent	2, 3
Design trade-offs	Each place a human decides is argued, not assumed; you can say what you chose not to automate and why	7
Governance and responsible AI	Audit trail complete; checkpoints named; bias and fairness check done and acted on; no invented facts reach a stakeholder	4, 5
Communication	Reports fit their audience; the panel can follow the system in 12 minutes; every member answers at least one question	8

Peer evaluation can also moderate an individual's share of the team marks where contribution was clearly unequal.

Rules

- **AI use is expected, and must be declared.** The design document lists which tools produced which parts of every deliverable, including the documents themselves.
- **Academic integrity applies as in the course policy.** Reusing another team's prompts, agents or evaluation results without attribution is collusion.
- **Sponsor questions go through the sponsor channels only.** Answers given to one team are posted to all teams.
- **Synthetic data only.** The data pack is fictional. Do not add real resident, employer or client information to any prompt, file or tool.
- **Team problems are raised early.** A member not contributing is raised with the instructor as soon as it happens, not first in the peer evaluation.

Appendix: data pack

The data pack simulates the project's first three months and is released on NTULearn with the first round of sponsor answers. All of it is synthetic. It is the only source your memory, risk and reporting modules may draw on, besides the brief and sponsor answers.

File	Format	Used by
Estate Development Committee minutes, 6 meetings	Text	Memory, reporting
Contractor monthly progress reports, 3 months	PDF	Memory, risk, reporting
Contract deliverables schedule	CSV	Change handling, risk
Seed risk register, inherited from the feasibility study	CSV	Risk
Resident consultation responses, about 200 comments	CSV	Governance (bias and fairness check)
Lessons learned from an earlier hub refurbishment	Text	Memory, risk
Grant milestone and tranche schedule	Text	Planning, reporting

The files do not always agree with each other or with the brief. Where they conflict, your system should say so and name both sources rather than pick one silently.